

Rat Olympic Final Report

I. Introduction and Experimental Objectives

For the “Rat Olympic” experiment, my partner and I aimed to use the principles of Operant Conditioning to teach a rat to navigate a tightrope structure. The specific objective was to shape the rat's behavior to walk down a tightrope, and then subsequently walk back across the tightrope, which was constructed using rope taped between two cardboard boxes (serving as platforms) on opposite ends of a large plastic tub, and secured to the top of both platforms with packaging tape as well. For our training purposes, we used two ropes; one was quite thick, and the other one was noticeably thinner. This ambitious task was chosen because the rat had previously demonstrated an elevated tendency to be highly physically active and interested in climbing the sides of its cage and up our arms, behaviors which were hypothesized to generalize quickly to climbing the rope.

For the purpose of positive reinforcement, my partner and I initially selected applesauce, thinking it would be sweet and avoid issues with perishable foods (like fresh fruit) or solid foods which our rat might find hard to eat or unfavorable due to texture (like rice, which was our very first thought). This reward system, however, quickly turned out to be a confounding variable. These were the only materials used in our experiment.

II. Detailed Description of Experimental Results

While the overall target of teaching the rat to cross the rope down and back was not fully achieved, the experiment successfully resulted in establishing the one-way rope crossing behavior. The rat learned to cross the rope unassisted from one platform to the other. This achievement was a result of rewarding successive approximations of the behavior, ultimately transforming a rat that would initially place no more than two paws on the rope into one capable of the full one-way traverse.

Due to the platforms being elevated and small in their size, our rat grew impatient with essentially nowhere to go. Our rat was learning faster than our anticipated schedule and was ready to explore the tightrope before day number two, which was the scheduled time for that behavior.

However, the final target behavior was not fully reached; we were unable to shape the rat to walk back down the rope to the starting platform, so that he began and ended on the same side. Furthermore, due to a procedural failure detailed below, my partner and I were unable to prompt the rat to engage in the rope crossing behavior. Consequently, during the final presentation of the trick to the Teaching Assistant and the rest of our class, my partner and I had to wait for the rat to initiate the behavior without external prompting. This reliance on the rat's self-initiation was particularly frustrating because the rat did not perform the crossing behavior frequently during training sessions, leading to the hypothesis that he became easily satiated or fatigued. Ultimately, our goal was unsuccessful because the end-goal of walking down and back across the tightrope was never accomplished.

III. Analysis of Confounding Variables and Obstacles (Why It Didn't Work)

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The shaping process was significantly slowed by several major obstacles and confounding variables, which ultimately contributed to the lack of time available to train the complex final target behavior (crossing back across the rope back to the starting platform). The goal was unsuccessful due to numerous confounding variables. However, looking back on our experiment I believe our primary issue was having an overly ambitious goal. Looking back on our experimental design, our results were doomed from the beginning, and our final target behavior was unlikely to ever occur.

A. Apparatus Instability (Physical Confounding Variable)

Possibly the most challenging obstacle encountered was the inherent lack of stability in the tightrope apparatus. The rope and cardboard platforms were secured only by packaging tape, rendering them completely lacking in stability. Additionally, the use of polyester rope made it difficult to pull the material completely taut, leading to slack on the rope. While the tightrope was able to support the rat's weight, the slack and lack of stability discouraged the rat from placing its full weight on the rope. My partner and I struggled for a long time to get him to place more than his front paws on the rope. The rat fell off or lost its balance several times during training, and this experience acted as a punishment. As a result, behaviors that involved placing more weight on the rope were less likely to occur following these punishing experiences, significantly slowing the shaping process.

If this experiment were to be repeated, stability issues could be addressed by using a different material, such as a wire material rather than polyester rope. Also, if possible, the rope and platforms might have been tied rather than just taped down to increase stability. I attempted to address this issue by adding more tape when switching to the thinner rope, but this did not increase stability as I had hoped.

B. Reinforcement Issues

The initial plan for positive reinforcement failed when it was discovered that the intended reinforcer, applesauce, was neither salient nor appetitive to the rat. The rat demonstrated disinterest and even actively moved its head away when the spoonful of applesauce was presented towards its face. This necessitated a deviation from the original plan. The issue was resolved by switching the reinforcement strategy to using sugar water in a dropper to reinforce behavior, and this worked much better as a positive reinforcement for behavior.

C. Procedural Failure: Lack of Target Training

A critical factor contributing to the failure of the full objective was the inability to successfully execute target training. The initial plan included training the rat to touch its nose to the spoon (or later, the dropper). This was included in our plan for the first few days of training. However, the training time was overly focused on encouraging the rat to approach the rope instead, and we did not teach the rat to touch its nose to the dropper first. We used the dropper to entice the rat to move towards or on the rope but did not shape a behavior in which he followed the dropper. Rather, we relied on the already existing inclination he had to move closer to the reward. Because we did not achieve successful target training, we were unable to prompt rope crossing behavior, and were also unable to prompt the rat to turn around and face the rope again after crossing. We were also unable to prompt him to cross from one platform to another, and when

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presenting his “trick” to the TA at the end of the experiment we had to wait for the rat to begin the behavior without being able to prompt rope crossing ourselves. This was particularly frustrating because I hypothesized that he was satiated or fatigued easily. He did not perform the crossing behavior often in a session.

D. Behavioral Obstacles: Escape Behavior and Biting

The rat made almost constant attempts to escape the training environment by jumping up onto the edges, which took away from time we spent training. For minutes at a time, I would watch him jump up onto the edges of the plastic bin and place him back on the platform only for him to jump back up on the edge of the plastic bin immediately after I placed him down. I sometimes inadvertently encouraged this problem behavior by sometimes allowing the rat to walk along the edge until it lost its balance or seemed like it was about to jump off, which may have taught the rat that if he jumped up, he would be rewarded with "rest time" at the top of the plastic bin.

If the experiment were repeated, the platforms should be made shorter and farther from the top of the plastic tub so that it was more difficult for him to jump up. Additionally, we should consistently pick up the rat and place it back on the platform every time he attempted to jump up on the edges of the plastic tub.

A related problem behavior that began around the fourth day of training was biting. This began to occur around the fourth day of Rat Olympics training. This was hypothesized to be a defensive behavior brought about by too much handling, necessitated by the rat's escape behaviors. This made training difficult because I had been using my hand to hold the slack on the rope when attempting to get him to cross, but I had to stop reaching into the training environment because it would result in biting. While the rat eventually learned to walk across without support from my hand, this issue slowed down the overall shaping process. I am unsure of how I could have better addressed this behavior without introducing positive punishment.

IV. Successful Achievement of Objectives

Despite the complexity of the obstacles, the successful establishment of the one-way rope crossing behavior was achieved through persistence, problem solving, and a heavy reliance on Operant Conditioning principles. In conclusion, despite never accomplishing our end-goal, we were indeed able to successfully train our rat to walk a tightrope, which is an quite impressive accomplishment itself.

1. Leveraging Existing Behavior: The experiment benefited from choosing a task that aligned with the rat's existing abilities, specifically its high level of physical activity and its elevated tendency to climb. We chose the rope for him because I had hoped that his previous behavior of climbing up my partner's and my arms as well as the edge of his cage would generalize quickly to climbing the rope, and I believe that because of his elevated tendency to be active and climb he was more equipped to complete the tightrope task than other rats in the lab.

2. Effective Reinforcement: The successful switch from the ineffective applesauce to the highly appetitive sugar water in a dropper ensured that the positive reinforcement successfully drove the desired behavior.

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3. Strategic Placement to Overcome Instability: To address the major issue of apparatus instability, we implemented a crucial strategy: I began placing the rat directly on the rope itself rather than on the platform. This made walking forward across the rope to the other platform the easiest way to get off the rope and obtain the reward. After starting this technique—where the rat walked halfway across the rope to a platform—the rat appeared more willing to walk fully across the rope beginning on one platform and ending on the other, resulting in the successful rope crossing behavior.

4. Flexible Training Schedule and Adaptability: Our training schedule was always forty minutes in total. For the first three days, twenty minutes of the training was split into two ten-minute periods, with twenty minutes of rest in between (always returning the rat to its home environment/cage for rest). On the fourth day of training, we modified our schedule to the following: a five-minute refresher period on yesterday's training, two ten-minute periods of training again, this time separated by a fifteen-minute period of rest. On the fifth day, the schedule changed again with the training time totaling fifty minutes now instead of the previous forty minutes allotted: a five-minute refresher period, a fifteen-minute training period, a ten-minute rest period, and ending with a twenty-minute training period. We maintained this schedule until the experiment's end. This flexibility allowed the partners to overcome confounding variables by changing training schedule and methods wherever we saw necessary.

5. Modeling Partner Behavior: Differences in training strategies were noted between partners, particularly in addressing escape behavior. My partner experienced less biting behavior, and so he was able to place his arm under the rope to provide stability and encourage target behavior. I think that this difference in biting behavior was due to strategies in addressing escape behavior. I often used my hands to block his escape, and sometimes blew gently on his face, which startled him and cause him to back off the edges of the plastic tub. I, unlike my partner, used his body and arms to discourage this behavior and never startled him by blowing on him. If I were to repeat this experiment, I would more closely match the behavior of both my partner and I.

Ultimately, through persistence and strategic adjustments, positive reinforcement successfully rewarded successive approximations of behavior, leading to the establishment of the one-way tightrope crossing. My partner and I were able to overcome the obstacles we faced in training and reach a version of our target behavior. The results were pleasing and demonstrated successful application of Operant Conditioning principles.